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COURSE NAME: ANTENNA AND WAVE PROPAATION

EX NO: 10B

ANALYSIS OF SKIP DISTANCE AND TROPOSPHERIC PROPAGATION USING MATLAB

DATE: 11/08/26

OBJECTIVES

i) To simulate Sky Wave Propagation and determine the Skip Distance for different ionospheric conditions using MATLAB.

ii) To analyze Tropospheric Propagation characteristics and determine the radio horizon distance for various antenna heights.

iii) To compare Skip Distance and Tropospheric Propagation coverage through graphical analysis and evaluate their impact on long-distance communication.

OBJECTIVE 1

ANALYSIS OF SKIP DISTANCE

PROCEDURE

1. Open MATLAB and create a new script file.
2. Define the ionospheric layer height.
3. Define the angle of incidence.
4. Calculate the skip distance using the propagation model.
5. Plot skip distance versus angle of incidence.
6. Analyze the effect of ionospheric height on skip distance.

FORMULA

Skip Distance:

$$D_s = 2h \tan \theta$$

Where:

- D_s = Skip Distance (km)
- **h** = Effective Ionospheric Height (km)
- **θ** = Angle of Incidence

RESULT

The skip distance for the given ionospheric conditions was successfully determined.

OBJECTIVE 2

ANALYSIS OF TROPOSPHERIC PROPAGATION

PROCEDURE

1. Define the transmitter antenna height and receiver antenna height.
2. Calculate the radio horizon distance.
3. Determine the tropospheric communication range.
4. Plot communication range versus antenna height.
5. Analyze the propagation characteristics.

FORMULA

Radio Horizon Distance:

$d_h = 4.12\sqrt{h}$

Tropospheric LOS Range:

$dLOS = 4.12(\sqrt{h_t} + \sqrt{h_r})$

Where:

- h_t = Transmitter antenna height (m)
- h_r = Receiver antenna height (m)

RESULT

The tropospheric communication range was successfully determined.

OBJECTIVE 3

COMPARATIVE ANALYSIS OF SKIP DISTANCE AND TROPOSPHERIC PROPAGATION

PROCEDURE

1. Simulate Sky Wave propagation.
2. Simulate Tropospheric propagation.
3. Plot the propagation distances.
4. Compare the communication coverage.
5. Analyze the propagation characteristics.
6. Interpret the results.

FORMULA

Coverage Ratio:

CR = D_s / dLOS

RESULT

The communication coverage of Sky Wave propagation was significantly larger than that of Tropospheric propagation.

OVERALL MATLAB PROGRAM

```
clc;

clear;

close all;

%% Objective 1 : Skip Distance Analysis

h = 250;                    % Ionospheric Height (km)

theta = 10:1:70;            % Angle of Incidence (degrees)

Skip_Distance = 2*h*tand(theta);

%% Objective 2 : Tropospheric Propagation

ht = 100;                   % Transmitter Height (m)

hr = 50;                    % Receiver Height (m)

LOS_Distance = 4.12*(sqrt(ht) + sqrt(hr));

%% Objective 3 : Comparative Analysis

Coverage_Ratio = Skip_Distance / LOS_Distance;

%% Display Results

fprintf('\nSKIP DISTANCE AND TROPOSPHERIC PROPAGATION ANALYSIS\n');

fprintf('----------------------------------------------------\n');

fprintf('Ionospheric Height = %.2f km\n', h);

fprintf('Skip Distance at 60° = %.2f km\n', ...

   2*h*tand(60));

fprintf('LOS Distance = %.2f km\n', ...

   LOS_Distance);

fprintf('Coverage Ratio = %.2f\n', ...

   (2*h*tand(60))/LOS_Distance);
```

```
%% Plot Results
figure('Name', ...
    'Skip Distance and Tropospheric Propagation', ...
    'NumberTitle','off');
% Skip Distance vs Angle
subplot(2,2,1);
plot(theta,Skip_Distance,'LineWidth',2);
grid on;
xlabel('Angle of Incidence (Degree)');
ylabel('Skip Distance (km)');
title('Skip Distance vs Angle');
% Tropospheric LOS Distance
subplot(2,2,2);
bar(LOS_Distance);
grid on;
title('Tropospheric LOS Distance');
ylabel('Distance (km)');
set(gca,'XTickLabel',{'LOS Range'});
% Coverage Ratio
subplot(2,2,3);
plot(theta,Coverage_Ratio,'LineWidth',2);
grid on;
xlabel('Angle of Incidence (Degree)');
ylabel('Coverage Ratio');
title('Coverage Ratio Analysis');
% Comparison
subplot(2,2,4);
plot(theta,Skip_Distance,'r','LineWidth',2);
hold on;
yline(LOS_Distance,'b--','LineWidth',2);
```

grid on;

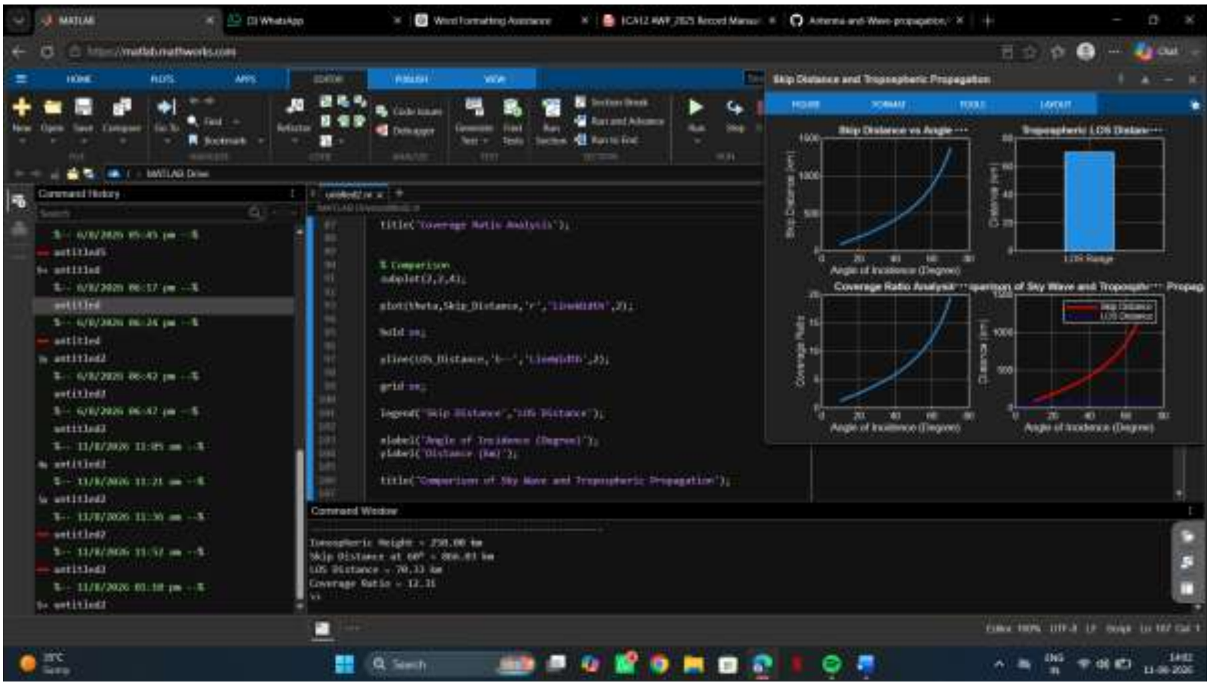
legend('Skip Distance','LOS Distance');

xlabel('Angle of Incidence (Degree)');

ylabel('Distance (km)');

title('Comparison of Sky Wave and Tropospheric Propagation');

OUTPUT



FINAL RESULT

The Skip Distance and Tropospheric Propagation characteristics were successfully simulated using MATLAB. The variation of Skip Distance with angle of incidence and the tropospheric LOS communication range were analyzed. The coverage ratio was also calculated, showing that Sky Wave propagation provides significantly greater communication coverage than Tropospheric propagation.